A working model of behavior leakage: a constant write, gated by context similarity

June 2026

1 Setup

Leakage: a behavior trained into a model under one context (a system prompt, persona, or ICL prefix) also appears under other, untrained contexts. We measure it as the trained−base change in the behavior's expression under a bystander context, on-policy, with a behavior-appropriate metric: marker-token log-probability at the end of the bystander's own response (#502/#489), judge-scored agreement rates for sycophancy (#509), misaligned-and-coherent rates for emergent misalignment (#521), taught-fact emission for fact implants (#500).

1.1 The picture in five claims

1. **Pretraining learns a dictionary of behaviors.** The model acquires a collection of behaviors b — sycophancy, marker emission, emergent misalignment — each carried by a direction in the residual stream at some layer. The direction plays two roles that should be named separately: *adding* a steering vector d_b elicits the behavior; *projecting* the state onto a read-out direction r_b measures it (the unembedding row for the marker; a probe or contrastive direction otherwise). In the idealized case one direction plays both roles ($d_b = r_b$); they can come apart, with a steering vector nearly orthogonal to the read-out that downstream circuitry converts (the router case, A1's relaxation in §2).

2. **The state under a context is a weighted sum over the dictionary.** A context c is summarized by a vector v_c: activations read while the model processes c, averaged over a probe set of user prompts. (Most generally the context representation is some function of the *KV cache* — that is all the response computation ever sees of c — and the pooled v_c is the cheapest member of that family; whatever the gate later misses traces back to what the pooling threw away.) The model's own context-to-state map f then puts the activations somewhere, and that somewhere decomposes approximately over the dictionary:

$$f(c) \;\approx\; \sum_b \alpha_b(c)\, d_b \;+\; \text{residual.}$$

   The point of the weights: the *associations* $c \to b$ that the model learned in pretraining ARE the coefficients $\alpha_b(c)$. Every context expresses many behaviors at once (its bundle); "the assistant is not a villain" is a statement about weights, not about the dictionary. (The decomposition is itself an idealization — dictionary directions are neither orthogonal nor a basis (superposition); it is literal for the marker at read-out, an empirical finding for EM, an open bet elsewhere.)

3. **Post-training edits the weight functions, never the dictionary.** Fine-tuning adds no behavior to the collection. In the lazy regime a small weight update cannot build new states or circuitry; it can only re-bind contexts to states that already exist — which is why a behavior is implantable only insofar as it is *elicitable* (prompting is how you locate the destination state that training will bind to). Shaping assistant behavior, marker implants, and EM induction are all edits of $\alpha_b(\cdot)$, differing in which weights move and for which contexts. Training b into a source context c' drags $f(c')$ toward v_b; leakage is how far other contexts' states move with it.

4. **Output is read out linearly, then squashed.** The state is not passed directly through a softmax. The chain is

   $$\text{state } f(c) \;\longrightarrow\; \text{logits } z_t = \langle W_U[t],\, f(c)\rangle \text{ per token } t \;\longrightarrow\; \text{softmax over the vocabulary} \;\longrightarrow\; \text{output},$$

   so the softmax is over *tokens*, not over behaviors. For the marker, behavior expression and a single logit coincide (z_{marker}), which is what makes it the clean testbed. For sycophancy, expression is a sequence-level property scored by a judge; every token still exits through the softmax, but the behavior is not one softmax entry. Because the softmax normalizes, behaviors compete at emission (marker vs end-of-turn) and log-probabilities compress near the ceiling — the reason logits are stored alongside log-probabilities.

5. **Two typed functions carry everything.** Keep them apart:

   - $f : c \mapsto f(c)$ — one state per context, no behavior argument. *This is what training edits.*
   - $A[b, c] = \langle r_b,\, f(c)\rangle$ — a scalar association table, rows indexed by behaviors, columns by contexts. *This is what experiments measure.*

   The behavior vector v_b is neither: it is the state f reaches when α_b dominates (read over response tokens) — where the model *is* when it is performing b. The division of labor matters: training targets the *state* v_b, observation projects onto the *direction* r_b, and nothing forces the two to align (§2).

1.2 Estimating the behavior vector

The behavior vector the update rule needs is the *post-training* state, unobservable before training, so estimating v_b from the base model is itself part of the proposal. The candidate estimators, in decreasing fidelity to the target read: (i) teacher-forced reads over the prospective training data's own completions; (ii) in-context demonstrations drawn from that data, read after the context and averaged over user prompts (the variant run so far); (iii) natural-language description conditioning. All three are f evaluated at a context that elicits the behavior, and each is validated by whether it matches the realized post-training state and forward-predicts leakage (P7–P8 in §3); the standing warning is a prompted marker direction unrelated to the realized training shift ($\cos \approx -0.03$, #521). Constructions (layer, read position, aggregation, estimator) are swept, not fixed; the experiments in §4 search over them.

Prompting stands in for fine-tuning in three places. Everything the formula consumes is read from the base model, which presupposes that prompting-derived objects track training-derived ones. The presupposition enters at three points of different strength, and they fail separately. *Mechanism*: an ICL prefix acts as a transient rank-one weight patch (A2's extension, §2), making prompting and fine-tuning the same operation at different persistence; full equivalence is open, so P7 claims only the behavioral form (in-context behavior forecasts the leakage map), which can hold even when the internal states do not match. *State identity*: the estimators above assume the state occupied during prompted performance of b is the state fine-tuning lands on. This is the most exposed of the three: P8 tests it directly, the standing warning above is one rung already failing, and the teacher-forced rung is the principled one, computed from the same forward passes the training gradients come from (A3's kernel relaxation). *Geometry persistence*: the gate evaluates base-model v_c to predict a model that has since been trained, assuming the edit leaves the measuring geometry intact; this is the lazy-regime requirement (§4), the one substitution no better estimator repairs. The update rule itself needs none of the three: with the realized $v_{b''}$ measured after training, equation (1) is descriptive. They are the price of prediction *before* training, which is the program's point, and most of §4 is implicitly testing them.

2 The simplest model

Assumptions. Each carries the relaxation that replaces it when it fails; the formula survives each in a weaker form.

A1 (behaviors are linear directions).

Adding a single residual-stream direction elicits a behavior [3, 4]; projections onto single directions track traits before generation [5]; emergent misalignment is predicted, controlled, and ablated through one direction [6, 7].

Relaxation: the write as a router (some behaviors are fetched, not resident). The write need not copy the behavior direction: a query shift can recruit pre-existing attention circuitry that fetches the behavior from the context's own values [8]. Cleanest case: a rank-1 EM adapter whose B vector has cosine 0.04 to the misalignment direction at its own layer, yet 98% of its effect ablates along that direction downstream [7]. The behavior stays one direction at read-out, but prediction P1 (§3) can fail by mechanism when fetched content differs across contexts.

A2 (edits are read/write pairs).

Weight updates read input directions and write output directions, with the rank-one outer product as the atomic unit [1]; low-rank adapters make this explicit [16, 2]. Narrow fine-tunes act as a near-constant steering write [8, 9], and the parameterization is irrelevant: at matched implant strength, LoRA and full fine-tuning leak identically (gap +0.00 nat, #514).

Extension: in-context edits. A transformer block provably converts an ICL prefix into a transient rank-one weight patch [17, 18]: the same write $\times$ gate, for the duration of the prompt. The gate already ranks leakage from ICL sources (#489), and in-context behavior forecasts fine-tuning outcomes (recovering $> 85\%$ of in-distribution corrections [19], anticipating generalization failures [22], compressing to a single direction [23]), though full ICL=fine-tuning equivalence remains open [20, 21]. Hence P7.

A3 (linearity, the extra assumption).

At one layer, $f(c) \approx M v_c$, and the edit acts at that layer.

Relaxation: kernel gate. For nonlinear f, a small update still acts to first order: $\Delta f(c) \approx K(c, c') \times \text{residual}$, with K the empirical tangent kernel [14, 15] (strictly a cross-kernel: read-out gradient at c against training gradient at c'). The gate becomes $g(c) = K(c, c')/K(c', c')$, the dot product of equation (1) its linear case; the proxy ladder runs cosine $\rightarrow$ distributional comparisons $\to$ exact kernel, with marginal gains so far (0.79 vs 0.76 marker; 0.50 vs 0.49 sycophancy). Under A1's routing the gate is not a similarity at all: attention-shaped, near-binary for discrete triggers, estimated from samples.

A4 (minimal edit).

Training implements the smallest weight change that fits the new pair: the rank-one key$\to$value write used by linear associative memories and ROME [10, 11, 12].

Relaxation: low-rank and multi-pair writes. Dropping rank-one, the write spans a subspace, $\Delta f(c) = U\,\mathrm{diag}(s)\,V^\top v_c$; P1 weakens to "shifts span r dimensions," and the SVD already in use measures r. With multiple pairs, the minimal edit solves a kernel system whose off-diagonal terms are the contrastive negatives, subtracting the common mode of near-parallel persona contexts (cosine ≈ 0.92); this is why positive-only implants leak near-uniformly while contrastive ones localize (99.6% source vs 11.7% bystander, #247/#329). Early stopping spectrally filters the solution toward raw similarity; the time-indexed predictor interpolating between raw and Gram-corrected similarity is untested.

From assumptions to formula. Fine-tune context c': training drags its state $v_{b'} = f(c')$ toward the behavior vector v_b, landing at an achieved state $v_{b''}$ (under early stopping the achieved state undershoots the target, so the write is measured, not assumed). A4 with A3 gives the edit: minimizing $\|\Delta M\|_F$ subject to $(M{+}\Delta M)\,v_{c'} = v_{b''}$ yields

$$M' = M + \frac{(v_{b''} - v_{b'})\,v_{c'}^\top}{\|v_{c'}\|^2} \qquad \Longrightarrow \qquad \Delta f(c) \;=\; \underbrace{(v_{b''} - v_{b'})}_{\text{write: } \textit{what} \text{ leaks}} \;\cdot\; \underbrace{\frac{\langle v_{c'}, v_c\rangle}{\|v_{c'}\|^2}}_{\text{gate: } \textit{who} \text{ leaks}} . \tag{1}$$

Written multiplicatively, the post-training map is $f'(c) = \big(M + (v_{b''} - v_{b'})\,v_{c'}^\top/\|v_{c'}\|^2\big)\,v_c$: a rank-one patch to the context-to-behavior map, with the normalization forced by the minimal-edit derivation. Each assumption does one job: A4 makes the edit rank-one (one write direction, one key), A3 makes the gate the activation inner product, A2 makes the same structure visible in the weights ($\Delta W \approx$ write $\otimes$ key), and A1 makes the write measurable, since the observed change is one slope per behavior times the gate: $\Delta z_b(c) = \langle r_b,\, v_{b''} - v_{b'}\rangle\, g(c)$. The softmax compresses log-probabilities near saturation and can reorder contexts there, which is why we store logits alongside log-probabilities.

One rank-one nudge to the association table. With the write $w = v_{b''} - v_{b'}$ and the table $A[b,c] = \langle r_b, f(c)\rangle$ of claim 5, equation (1) compresses to

$$\Delta A[b,c] \;=\; \underbrace{\langle r_b,\, w\rangle}_{\text{behavior factor}} \;\cdot\; \underbrace{g(c)}_{\text{context factor}}$$

— a rank-one update of the association table itself. The intended edit is one cell (b, c'); the minimal edit cannot touch one cell, it smears along both axes. *Down the columns* (across contexts): every context moves in proportion to its gate — that smear is leakage (P2). *Across the rows* (across behaviors): every behavior moves in proportion to its slope $\langle r_b, w\rangle$ — that smear is cross-behavior transfer (P6). Leakage and cross-behavior transfer are the same event seen along the two axes of the table, and leakage gets a clean definition in dictionary coordinates: off-target table movement — coefficients $\alpha_b(c)$ changing for behavior–context pairs the fine-tune never targeted. The intended edit is one cell; everything else that moves is generalization nobody asked for.

The write is not the read-out. Collapsing v_b and r_b into one "behavior direction" would smuggle in an assumption the formula does not make: that the network between the edit layer and the output acts as the identity on the write, so the direction pushed in is the direction read out. Kept separate, the downstream network is free to rotate, amplify, or route the write, and the anomalies observed so far live exactly in that gap. Each has the form "write and read-out nearly orthogonal, yet the behavior moved" — a sentence with two direction-valued slots, which a one-object model cannot state; it sees only an unexplained slope. Three are on record: the prompted marker direction against the realized training shift ($\cos \approx -0.03$, #521; an estimator failure, P8); the rank-1 adapter whose write has cosine 0.04 to the misalignment direction yet loses 98% of its effect to ablation along that direction downstream [7] (routing, not copying; A1's relaxation); and the P1 split (§4), where the marker and EM arms are indistinguishable at the read-out — each a table of per-context behavior shifts — and differ only in activation space, in whether the shift matrix is rank-one. The split should not be oversold either: for a single behavior the observable compresses to $\Delta z_b(c) = s_b\, g(c)$ with s_b a fitted scalar, ranking who leaks needs no v_b at all, and the minimal model reads fairly as slope × gate. The two objects earn their keep in the vector-valued predictions — P1 (one write direction), P3′ (the write's common-mode fraction: "not weaker, narrower" has no scalar reading), P6 (one write projected onto many read-outs), P8 (r_b is directly observable while v_b must be estimated) — the predictions that make this a mechanism rather than a regression. The objects are also different kinds: v_b is a state with position and norm (P3's source residual $\|v_{b''} - v_{b'}\|$ requires points that subtract), r_b a direction at the output interface.

The edit has two channels. Writing the linear rung as $f(c) = M v_c$, a fine-tune can change the response state in two places:

$$\Delta f(c) \;=\; \underbrace{\Delta M\, v_c}_{\text{re-bind the map}} \;+\; \underbrace{M\, \Delta v_c}_{\text{move the representation}} \;+\; \Delta M\, \Delta v_c,$$

and nothing confines the update to M: v_c is computed by the same network the adapter modifies (an all-linear adapter touches every layer below the read), so the context representation itself is fair game. Equation (1) is the first channel alone — A3 places the edit at the read layer, and geometry persistence (§1) is exactly the assumption that the second channel vanishes. A live second channel has so far been filed wholesale as the rich regime (§4), a verdict of unpredictability; written as a decomposition it becomes a measurement instead, and both outcomes are already on record: tripling implant depth left the context geometry unchanged (#538; $\Delta v_c \approx 0$), while EM training rotated the assistant axis 38–53° and collapsed persona discrimination (#125; a Δv_c no frozen-v_c predictor survives). The channels separate on stored artifacts: re-read each trained adapter's context summaries against base (Δv_c directly), fit the rank-one patch on frozen base summaries (the $\Delta M\, v_c$ share), and assign the residual to the moved representations. That read is also the cheapest candidate account of the P1 tension (§4): the marker and EM arms may put their edits in different channels, and per-arm channel shares on the stored #521/#551 shift tensors would say so before any new training.

Contexts encode behaviors. A context and a behavior are the same kind of object here: both are activation states, and a persona prompt ("you are a villain") is a point whose projection profile onto the read-out directions, $\{\langle r_b, f(c)\rangle\}_b$, is the behavior bundle it elicits without any training. The model uses this in three places: the source residual (an instructed source already encodes b, so the write is small; P3), the bystander prior (a context that already encodes b expresses more of it at the same push; P3), and the estimator ladder itself (v_b is read from behavior-eliciting contexts).

What it does *not* license is the write $\times$ gate factorization treating behavior and context as separate axes: for persona-like contexts the gate $\langle v_{c'}, v_c \rangle$ mixes prompt similarity with behavior-bundle overlap, and the two can come apart: an instruction context can encode the trained behavior while sitting geometrically far from the source. That is the regime where geometry collapsed and the base prior carried the signal (#532), so the open test is to vary behavior overlap at fixed context similarity (matched-geometry bystander panels with different base priors on b) and ask whether the gate needs a behavior-overlap term of its own. One concrete candidate for that term conditions the similarity measure on the behavior itself: score a context pair by the divergence between their output distributions *on the material being taught* — e.g. the JS divergence over the taught answers, teacher-forced on the training questions — rather than over generic probe outputs, so two contexts count as close exactly when they agree about b, and the gate inherits the behavior dependence the raw dot product lacks. Its scalar limit is the target context's log-probability of the taught answers, which re-derives the base-prior predictor (P3, #532) as a gate term rather than a read-out correction, and it sharpens the matched-geometry test: where the generic-question measure collapsed on instruction contexts, the taught-answer-conditioned one should keep ranking leakage if the behavior-overlap term is real.

Cast of characters.

Object	Type	Role	Moved by training?
v_c	vector	context key (input side), read while processing c	no (base-model object)
f	map	contexts $\to$ states; holds the associations	**yes — the only thing edited**
$f(c)$	state	where the model sits under c	moves when f moves
$\alpha_b(c)$	scalar	association strength of $c \to b$ (the weights)	moves (this *is* the edit)
d_b	direction	steering: add to elicit b	no (dictionary)
r_b	direction	read-out: project to measure b	no (and directly observable)
v_b	state	where the model is when performing b (target)	no (dose-independent)
$v_{b'}$, $v_{b''}$	states	source state before / achieved after training	$v_{b''}$ is the outcome
$w = v_{b''} - v_{b'}$	vector	the write: *what* leaks	created by training
$g(c)$	scalar	the gate: *who* leaks	no (computed from base v_c)

3 Predictions

P0 (weights decompose).
The top singular vectors of the trained ΔW are the key and the write. Under contrastive training the key moves from raw $v_{c'}$ toward a source-minus-negatives contrast direction as dose grows.

P1 (one write direction).
Per-bystander activation shifts at the read-out all point along $(v_{b''} - v_{b'})$; an SVD of the shift matrix is near rank-one with a top component matching the behavior delta.

P2 (the gate is base-model context similarity).
Some fixed kernel on base-model context summaries sets who leaks, and it transfers across held-out panels. On the linear rung it is the dot product, hence asymmetric: $c' \to c$ scales as $\cos\theta \cdot \|v_c\|/\|v_{c'}\|$.

P3 (the behavior's prior matters, through three channels).
Source side: the write scales with the residual $\|v_{b''} - v_{b'}\|$, so a source that already performs

the behavior produces little leakage despite high similarity. Bystander side: the prior enters through read-out compression (measurement) and possibly through the gate itself (mechanism); the two separate in logit space, but only with the exact gate rather than a proxy.

P3′ (the source prior shapes the write's direction, not just its size).
If the source already carries the behavior's shared component, little of it needs writing: the residual concentrates in source-specific subspace, so the source prior predicts the write's *common-mode fraction* (its projection onto the shared / mean-bystander shift direction), not its norm. This is the configuration #541 observed — the high-prior teachers express the implant *most* on themselves (97/94% self-assertion vs the leaky low-prior teacher's 72%) while leaking least ("not weaker, narrower") — which the magnitude-only reading of P3 cannot produce: a uniformly small write would weaken expression everywhere, including source-self.

P4 (additivity).
Two implants trained jointly leak like the sum of their singletons, with deviations set by the sources' mutual similarity (the activation-space analogue of task arithmetic [13]).

P5 (layer locality).
The gate predicts best at the layers where the behavior is computed; scoring this needs an independent localization (patching or probing), since scoring by predictor quality is circular. The layer is behavior-specific, not universal: the marker reads best late (19–24), sycophancy early (1–8).

P6 (cross-behavior transfer).
The same write moves other behaviors in proportion to its projection onto their read-out directions, extending the rule from context generalization to behavior generalization.

P7 (ICL proxy).
The base model's in-context behavior on the training mix forward-predicts the post-training leakage map (see A2's extension for why).

P8 (the estimator is valid).
The base-model estimates of v_b (teacher-forced, demonstration-conditioned, description-conditioned) agree with each other and with the realized post-training state $v_{b''}$ at the read-out layers. If they do not, formula predictions made with the estimated v_b fail in a way that substituting the realized $v_{b''}$ repairs, and the estimation problem, not the update rule, is what failed.

No single miss refutes the model, since each assumption carries a relaxation (§2); the conjunction does. With the gate computed exactly and the write measured on a rig-clean implant, a failure of write $\times$ gate to rank leakage on a held-out panel leaves no version standing.

4 Experiments

	Experiment	Status
P0	SVD stored adapters; compare singular vectors to keys and measured writes; cross-compare [8]'s per-trigger backdoor vectors	Not yet run; free on existing artifacts.
P1	SVD per-context activation shifts, marker implant vs EM SFT (#521, controls #551)	Split: EM shifts all 14 contexts along one seed-stable direction (cos 0.96–0.98); the marker implant does not, and its trained context is *least* aligned. Benign SFT also gives one direction (#552, running); positive-only retrain tests the rig confound (#561, running).
P2	Correlate base-model similarity with measured leakage across source×bystander panels (#502, #489, #509)	Similarity ranks marker leakage ($\lvert\rho\rvert \approx 0.76$–$0.79$, layers 19–24) and extends to ICL/multi-turn contexts, but only ordinally (out-of-fold $R^2 \approx 0$, single-seed); structure does not transfer across panels (#553); the fact arm reversed sign ($\lvert\rho\rvert = 0.49$) while the base prior predicted correctly (+0.27, #500), an in-regime miss.
P3	Instructed-source break case; regress logit shifts on gate proxy + bystander base logit (#500, #531, #532)	The bystander's own prior is the most consistently supported predictor and beats geometry on instruction contexts (#532); the measurement-vs-mechanism split awaits the exact gate.
P3′	Decompose measured writes: project per-context activation shifts onto shared (mean-bystander) vs source-specific components; regress the common-mode fraction on source prior (#541's nine fact adapters span teacher prior; #518's refusal/EM adapters extend across behaviors)	Not yet run; activation extraction on existing adapters only, no new training.
P4	Joint two-source training vs singleton sum (#520, #527, #538)	Not yet diagnostic: singleton shifts are themselves near rank-one, so the additivity cosine (0.99) is trivially high at every tested dose.
P5	Layer sweep of predictor quality per behavior + independent localization	Marker predicts best late (19–24), sycophancy early (1–8); consistent but unconfirmed without the localization step.
P6	Behavior-generalization testbed (#545); context testbed (#537); formula-as-predictor (#510)	#537 running; #545, #510 queued.
P7	ICL-vs-FT equivalence, persona-framed (#491): same K examples in-context vs fine-tuned; ICL-conditioned base responses as a leakage predictor	Queued; the similarity gate already extends to ICL source contexts (#489).
P8	Estimator bake-off: score the three v_b estimators against realized post-training states on stored adapters (#493 extraction engine, #521 shift tensors)	Not yet run; free on existing artifacts. Warning already on record: the prompted marker direction is unrelated to the realized shift (cos ≈ -0.03, #521).

The open headline experiment is a dose titration: implant at a non-saturated anchor (#448), measure expression change, activation shift, and gate proxy per held-out context, on-policy (#432→#456), at several doses across the 10–25-step cliff (#139), with at least 3 seeds. At low dose the shift matrix should be near rank-one with similarity-graded magnitude; across the cliff the direction should rotate and the prediction collapse. The P1 split already complicates the low-dose half, so the titration doubles as its diagnosis.

What cannot be relaxed. Every version assumes the update is small in weight/feature space, though not in behavior: read-outs amplify, so a lazy edit can flip emission discretely or induce EM through nine rank-1 adapters [7], and tripling implant depth left the edit geometry unchanged (#538). If training moves the features themselves (the rich regime), no base-model predictor can work — geometry persistence fails, and the other two substitutions of §1 fail with it — and the historical EM predictor failure looks like that regime ($\rho = 0.09$ prose-only #458; assistant-axis rotation 38–53°, discrimination collapse, #125). The current tension: the marker arm passes the gate test but fails the one-direction test, while EM does the reverse; #552 and #561 decide whether the rig or the regime explains it. The durable question under all versions is the same: how far is fine-tuning generalization predictable from base-model activation geometry, before training?

The best-effort question. The control-side twin of that question commits to no gate at all: out of every possible embedding of contexts — nonlinear, any function of the KV cache — pick the one most discriminatory for the behavior at hand, suppose fine-tuning could gate on it, and ask whether leakage persists.

- *If leakage is inevitable even at best effort, the reason has to be named.* The lazy-regime candidate is expressivity: a small weight update can only gate on directions the base model already represents (approximately) linearly; if separating the source from the bystanders requires a nonlinear function of the KV cache that the base model never linearized, no small edit can read it. Leakage is then forced by the editing mechanism rather than by information — inevitable leakage = the gap between the best discriminator and the best discriminator expressible as a linear gate on frozen features.
- *If leakage is evitable, the answer should be constructive: exhibit the edit.* The multi-pair contrastive edit (A4's relaxation) is the linear-case candidate: solving the kernel system subtracts the common mode of near-parallel keys (cosine ≈ 0.92) and zeroes the gate on the enumerated negatives; its known limitation is exactly the right test — protection extends only to (roughly) the span of the negatives trained against, so leakage to contexts *off* the panel measures how far best effort reaches.

Under linear embeddings the dichotomy reduces to geometry: a rank-one gate is $g(c) = \langle k, v_c \rangle$ for an effective key k, and zeroing every bystander while still firing on the source is possible exactly when $v_{c'}$ has a usable component outside the bystanders' span — leakage is evitable iff the source is linearly separable from the bystander distribution in key space, with contrastive negatives an estimator of the discriminant direction. The open part is the continuum: "all contexts" is not an enumerable panel, and whether a discriminant fitted against K negatives generalizes off-panel is an empirical question, not a theorem. Prediction (how far is generalization forecastable) and control (is off-target generalization avoidable) are the two faces of the same geometry.

References

[1] N. Elhage et al. A mathematical framework for transformer circuits. *Anthropic (transformer-circuits.pub)*, 2021.

[2] J. Ward et al. Rank-1 LoRAs encode interpretable reasoning signals. arXiv:2511.06739, 2025.

[3] A. Turner et al. Activation addition: steering language models without optimization. arXiv:2308.10248, 2023.

[4] N. Panickssery et al. Steering Llama 2 via contrastive activation addition. arXiv:2312.06681, 2023.

[5] R. Chen, A. Arditi, H. Sleight, O. Evans, J. Lindsey. Persona vectors: monitoring and controlling character traits in language models. arXiv:2507.21509, 2025.

[6] M. Wang et al. Persona features control emergent misalignment. arXiv:2506.19823, 2025.

[7] A. Soligo, E. Turner, S. Rajamanoharan, N. Nanda. Convergent linear representations of emergent misalignment. arXiv:2506.11618, 2025.

[8] J. Engels et al. Simple mechanistic explanations for out-of-context reasoning. arXiv:2507.08218, 2025.

[9] J. Minder et al. Narrow finetuning leaves clearly readable traces. arXiv:2510.13900, 2025.

[10] T. Kohonen. Correlation matrix memories. *IEEE Trans. Computers*, 1972.

[11] J. A. Anderson. A simple neural network generating an interactive memory. *Mathematical Biosciences*, 1972.

[12] K. Meng, D. Bau, A. Andonian, Y. Belinkov. Locating and editing factual associations in GPT. *NeurIPS*, 2022.

[13] G. Ilharco et al. Editing models with task arithmetic. *ICLR*, 2023.

[14] A. Jacot, F. Gabriel, C. Hongler. Neural tangent kernel: convergence and generalization in neural networks. *NeurIPS*, 2018.

[15] S. Malladi, A. Wettig, D. Yu, D. Chen, S. Arora. A kernel-based view of language model fine-tuning. *ICML*, 2023.

[16] E. Hu et al. LoRA: low-rank adaptation of large language models. *ICLR*, 2022.

[17] B. Dherin et al. Learning without training: the implicit dynamics of in-context learning. arXiv:2507.16003, 2025.

[18] Goldwaser et al. arXiv:2511.17864, 2025 (extends [17] to gated, bias-free, RMSNorm transformer blocks).

[19] D. Dai et al. Why can GPT learn in-context? Language models implicitly perform gradient descent as meta-optimizers. arXiv:2212.10559, 2022.

[20] L. Shen, A. Mishra, D. Khashabi. Do pretrained transformers really learn in-context by gradient descent? arXiv:2310.08540, 2023 (ICML 2024).

[21] G. Deutch et al. In-context learning and gradient descent revisited. arXiv:2311.07772, 2023 (NAACL 2024).

[22] A. Lampinen et al. On the generalization of language models from in-context learning and finetuning: a controlled study. arXiv:2505.00661, 2025.

[23] S. Liu et al. In-context vectors: making in-context learning more effective and controllable through latent space steering. arXiv:2311.06668, 2023 (ICML 2024).